

Frames with diagonal frame operator: exact operator and perturbation parametrizations

Candidate full solution to two explicit questions in arXiv:2309.06331

11 August 2026

Status: candidate full solution, likely valid. Fix the orthonormal basis with respect to which *diagonal* is meant. If a frame has frame operator S , then all bounded operators producing a diagonal-frame-operator frame are exactly

$$M = D^{1/2}CS^{-1/2},$$

where D is positive, invertible, and diagonal and C is a coisometry. All additive perturbations are likewise exactly

$$d_j = D^{1/2}Ce_j - v_j.$$

These formulas hold in finite and infinite dimensions. The packet also settles the source's canonical-dual boundedness question and gives a sharp minimax characterization of componentwise perturbation radii that preserve all frames with a prescribed lower bound.

1 The source questions

Let $\{v_j\}_{j \in J}$ be a frame in a real or complex Hilbert space. The source asks which operators M make $\{Mv_j\}$ a frame whose frame operator is diagonal, and equivalently which additive perturbations $\{d_j\}$ make $\{v_j + d_j\}$ such a frame [1, p. 11]. The full source page is reproduced below.

Thus we answer in the affirmative: a tight frame can encounter erasures so that the resulting set is still a tight frame, but that occurs if and only if the removed vectors form a tight frame themselves.

We have provided a complete characterization of a tight frame encountering erasures and whether the resulting set after the erasures is a tight frame or not:

Corollary 4.3.5.

- *If a tight frame for $\mathbb{R}^n$ encounters $p < n$ erasures, the resulting set is never a tight frame.*
- *If a tight frame for $\mathbb{R}^n$ encounters $p \geq n$ erasures, the resulting set is a tight frame if and only if the set of vectors removed form a tight frame themselves.*

The condition number of a frame corresponds to the classical condition number of the frame operator matrix which is a measure of the effect of perturbations on the inverse. A low condition number is desired for purposes of numerical stability in the reconstruction formula which makes use of the inverse of the frame operator matrix. Changes in the condition number after erasures is of interest and [8] states in Proposition 2.5 that for a unit norm tight frame the condition number always increases after a single erasure. But we can now extend Proposition 2.5 and drop the requirement that the tight frame must be unit norm:

Proposition 4.4. *The condition number of a tight frame always increases after a single erasure*

Proof. A single erasure on a tight frame, which has condition number 1, leaves the resulting set not being a tight frame with a condition number greater than 1. $\square$

This raises an important question: what is the new condition number of a tight frame that has had some frame elements removed? Proposition 2.10 in [8] establishes a worst case and best case of the new condition number after a single erasure but it is worthwhile to explore this problem further in more generality.

5. OPEN PROBLEMS AND REMARKS

The representation formula (2.1) requires the inversion of the frame operator matrix. The optimal scenario from a computational perspective is encountered with tight frames in which the frame operator is a multiple of the identity and hence the easiest to invert. The next best case would be having a diagonal frame operator as they are also easy to invert and thus offer benefits similar to tight frames. In [9] the following question is posed as a possible research direction: Given a frame $\{v_i\}$, what are the operators M for which $\{Mv_i\}$ is a frame whose frame operator is a diagonal matrix? This question can also be approached from the perspective of additive perturbations: Given a frame $\{v_i\}$, what are the perturbations $\{d_i\}$ for which $\{v_i + d_i\}$ is a frame whose frame operator is a diagonal matrix?

We show a simple example of a perturbation that transforms a frame in $\mathbb{R}^2$ into a frame with a diagonal frame operator.

We denote with $e_1 = (1, 0), e_2 = (0, 1)$ the vectors of the canonical basis of $\mathbb{R}^2$.

The next page asks how far a frame must be perturbed to obtain a diagonal frame operator. It then turns to infinite dimensions, observing that the boundedness of the canonical dual is not immediately obvious, and asks for the perturbation radii in the Krein–Milman–Rutman setting that preserve frames [1, p. 12].

Proposition 5.1. *If $\{v_j\}_{j=1}^k$ is a nonzero frame in $\mathbb{R}^2$, then there exists $\epsilon \in \mathbb{R}$ such that the frame operator of $\{v_j\}_{j=1}^{k-1} \cup \{v_k + \epsilon e_j\}$ is diagonal for some $j \in \{1, 2\}$.*

Proof. Let $T = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ be the synthesis operator for $\{v_j\}_{j=1}^k$. Here u_1, u_2 are the rows of the matrix T . Let $v_s(i)$ be the entry in T with largest absolute value i.e. maximum $|v_j(i)|$. Reorder the frame elements where this is the last vector v_k .

Let

$$\epsilon = -\frac{\langle u_1, u_2 \rangle}{v_k(i)}$$

and let $T' = \begin{bmatrix} u'_1 \\ u'_2 \end{bmatrix}$ be the synthesis operator for $\{v_j\}_{j=1}^{k-1} \cup \{v_k + \epsilon e_i\}$. Thus,

$$\langle u'_1, u'_2 \rangle = \langle u_1, u_2 \rangle + \epsilon v_k(i) = 0.$$

Hence, the frame operator $T'(T')^*$ is diagonal. $\square$

With this generalization, we can rephrase one of the questions that we posed in the introduction regarding frame stability in terms of diagonal frames:

(2*) How much does a frame need to be perturbed to become a frame with a diagonal frame operator?

5.1. Infinite Dimensional Frames. To conclude this paper, we give some remarks about extending our main results to infinite dimensional Hilbert spaces.

The construction given in Theorem 1.2 relies on the following fact in finite dimensions:

$$\max_{j \in \{1, \dots, k\}} \|S^{-1}v_j\| < \infty.$$

In other words, the canonical dual frame is always bounded. However, in infinite dimensional spaces, it is not immediately obvious whether this is true.

The construction given in Theorem 1.3 needs the frame operator to have finitely many eigenvalues. In infinite dimensions, the theorem holds for frames with frame operators having finite spectrum. However, most frames operators do not have this property, so more powerful tools will need to be used.

As mentioned in the introduction, a corollary to the proof of the Krein-Milman-Rutman theorem forbids an infinite dimension analog to Proposition 4.1. We leave as an open problem to characterize the additive perturbations $\{\epsilon_n\}_{n \in \mathbb{N}}$ in the Krein-Milman-Rutman theorem that preserve frames.

We state a simple convexity property of perturbed frames which could allude to thinking of frames topologically:

Proposition 5.2. *Suppose $\{x_n\}_{n \in \mathbb{N}}$ is a frame in $\mathcal{H}$ with frame bounds A, B . Moreover, suppose $\{y_n\}_{n \in \mathbb{N}}$ is a sequence in $\mathbb{H}$ such that $\{x_n\}_{n \in \mathbb{N}}$ and $\{y_n\}_{n \in \mathbb{N}}$ satisfy (2.3) with $\lambda = 0$ and $0 \leq \mu < \sqrt{A}$. Let $\tau < \frac{\sqrt{A}}{\mu}$. Then, for every sequence of complex numbers $\{t_n\}_{n \in \mathbb{N}}$ with $|t_n| \leq \tau$,*

$$\{(1 - t_n)x_n + t_n y_n\}_{n \in \mathbb{N}}$$

2 Setup and proof intuition

Let $T : \ell^2(J) \rightarrow \mathcal{H}$ be the synthesis operator, $Te_j = v_j$. The frame operator is $S = TT^*$, and the frame condition is exactly that S is positive and boundedly invertible. There are then only two identities to use:

$$\text{frame operator of } \{Mv_j\} = MSM^*, \quad \text{frame operator of } \{We_j\} = WW^*.$$

Factoring a desired positive diagonal operator D turns either identity into $CC^* = I$. Thus the apparently nonlinear classification is precisely the elementary parametrization of factorizations of D

through a coisometry.

Throughout, diagonal operators are diagonal relative to a fixed orthonormal basis of $\mathcal{H}$. A coisometry $C : \mathcal{K} \rightarrow \mathcal{H}$ means $CC^* = I_{\mathcal{H}}$. Positive invertibility of a diagonal operator includes uniform lower and upper bounds on its diagonal entries.

3 Complete operator classification

Theorem 1 (operator parametrization). *Let $\{v_j\}_{j \in J}$ be a frame for $\mathcal{H}$, with frame operator S . For $M \in \mathcal{B}(\mathcal{H})$, the following are equivalent.*

1. *The family $\{Mv_j\}_{j \in J}$ is a frame and its frame operator is diagonal.*
2. *There are a positive invertible diagonal operator D on $\mathcal{H}$ and a coisometry $C : \mathcal{H} \rightarrow \mathcal{H}$ such that*

$$M = D^{1/2}CS^{-1/2}.$$

For a specified resulting frame operator D , this describes all such M . In finite dimensions C is unitary (orthogonal over the reals).

Proof. The transformed synthesis operator is MT , so its frame operator is MSM^* . Suppose first that this is a positive invertible diagonal operator D , and define

$$C = D^{-1/2}MS^{1/2}.$$

Then

$$CC^* = D^{-1/2}MSM^*D^{-1/2} = I,$$

and rearrangement gives the asserted formula for M .

Conversely, if $M = D^{1/2}CS^{-1/2}$ and $CC^* = I$, then

$$MSM^* = D^{1/2}CC^*D^{1/2} = D.$$

This operator is positive and boundedly invertible, so $\{Mv_j\}$ is a frame with diagonal frame operator D . If $\mathcal{H}$ is finite-dimensional, a square coisometry is unitary. $\square$

The coisometry cannot in general be replaced by a unitary in infinite dimensions: the backward shift on $\ell^2(\mathbb{N})$ is a noninjective coisometry. The formula correctly allows M to be surjective without being injective.

4 Complete additive classification

Theorem 2 (additive parametrization). *Let $T : \ell^2(J) \rightarrow \mathcal{H}$ synthesize the frame $\{v_j\}_{j \in J}$. A sequence $\{d_j\}_{j \in J} \subset \mathcal{H}$ has the property that $\{v_j + d_j\}_{j \in J}$ is a frame with diagonal frame operator if and only if there are a positive invertible diagonal $D \in \mathcal{B}(\mathcal{H})$ and a coisometry $C : \ell^2(J) \rightarrow \mathcal{H}$ such that*

$$d_j = D^{1/2}Ce_j - v_j \quad (j \in J).$$

The resulting frame operator is exactly D .

Proof. Put $w_j = v_j + d_j$. If $\{w_j\}$ is a frame, its synthesis operator $W : \ell^2(J) \rightarrow \mathcal{H}$ is bounded. Its frame operator is WW^* . If $WW^* = D$ is positive, invertible, and diagonal, then

$$C = D^{-1/2}W \quad \text{satisfies} \quad CC^* = D^{-1/2}WW^*D^{-1/2} = I.$$

Thus $W = D^{1/2}C$, and evaluation at e_j gives the displayed formula. The converse follows at once from $WW^* = D^{1/2}CC^*D^{1/2} = D$. $\square$

This also yields an entrywise criterion. Let $\{h_k\}_{k \in K}$ be the fixed orthonormal basis of $\mathcal{H}$, and put

$$r_k = W^*h_k = (\langle h_k, w_j \rangle)_{j \in J} \in \ell^2(J).$$

Then WW^* is diagonal exactly when the rows r_k are pairwise orthogonal. The family is a frame exactly when, in addition,

$$0 < \inf_{k \in K} \|r_k\|^2 \leq \sup_{k \in K} \|r_k\|^2 < \infty.$$

Hence Theorem 2 is equivalently a complete orthogonal-row characterization.

Corollary 3 (an explicit diagonalizing perturbation). *Every frame can be additively perturbed to a Parseval frame, hence to one with diagonal frame operator, by*

$$d_j = (S^{-1/2} - I)v_j.$$

If $E = (S^{-1/2} - I)T$ is the synthesis perturbation, then

$$EE^* = (S^{1/2} - I)^2, \quad \|E\| = \|S^{1/2} - I\|.$$

In finite dimension the total squared displacement is

$$\sum_j \|d_j\|^2 = \text{tr}(S) - 2 \text{tr}(S^{1/2}) + \dim \mathcal{H}.$$

Proof. The new synthesis operator is $W = S^{-1/2}T$, and $WW^* = I$. Also $EE^* = (S^{-1/2} - I)S(S^{-1/2} - I) = (S^{1/2} - I)^2$. Taking operator norms proves the second formula, and taking the trace proves the finite-dimensional identity. $\square$

The corollary supplies an exact cost for this canonical construction; it does not assert that this is the closest frame among all frames having an arbitrary diagonal frame operator.

5 Two infinite-dimensional consequences

Proposition 4 (the canonical dual is uniformly bounded). *If $\{v_j\}$ has frame bounds A, B , then its canonical dual $\{S^{-1}v_j\}$ is a frame with bounds $1/B, 1/A$. Consequently*

$$\sup_j \|S^{-1}v_j\| \leq A^{-1/2}.$$

This estimate is sharp.

Proof. The canonical dual has synthesis operator $S^{-1}T$ and frame operator $S^{-1}TT^*S^{-1} = S^{-1}$. Since $AI \leq S \leq BI$, functional calculus gives $B^{-1}I \leq S^{-1} \leq A^{-1}I$. A Bessel family with bound $1/A$ has each vector of norm at most $A^{-1/2}$. Equality holds for the scaled orthonormal basis $v_j = \sqrt{A}e_j$. $\square$

The next theorem gives a sharp universal version of the source's perturbation-radius question. It separates a guarantee based only on the original lower frame bound and the radii from the subtler question for one fixed frame.

Theorem 5 (sharp universal componentwise radii). *Fix $A > 0$ and nonnegative finite radii $\varepsilon = (\varepsilon_j)_{j \in J}$, where J is finite or countable. The following are equivalent.*

1. *For every Hilbert-space frame $\{v_j\}_{j \in J}$ with lower frame bound at least A , every sequence satisfying $\|d_j\| \leq \varepsilon_j$ leaves $\{v_j + d_j\}$ a frame.*
2. *$\varepsilon \in \ell^2(J)$ and $\|\varepsilon\|_2 < \sqrt{A}$.*

When (2) holds and the original upper bound is B , every such perturbed frame has bounds

$$(\sqrt{A} - \|\varepsilon\|_2)^2, \quad (\sqrt{B} + \|\varepsilon\|_2)^2.$$

Proof. Let $E : \ell^2(J) \rightarrow \mathcal{H}$ be the synthesis perturbation, $Ee_j = d_j$. If $\delta = \|\varepsilon\|_2 < \infty$, then for finitely supported c ,

$$\|Ec\| \leq \sum_j |c_j| \varepsilon_j \leq \delta \|c\|_2,$$

so $\|E\| \leq \delta$. For the original synthesis operator T , the standard synthesis form of the frame inequalities gives

$$\|(T + E)^*x\|_2 \geq (\sqrt{A} - \delta)\|x\|, \quad \|(T + E)^*x\|_2 \leq (\sqrt{B} + \delta)\|x\|.$$

This proves sufficiency and the asserted bounds.

For necessity, first suppose $\varepsilon \notin \ell^2(J)$. Given any frame $\{v_j\}$, choose a unit vector h and set $d_j = \varepsilon_j h$. The scalar sequence $(\langle h, v_j \rangle)_j$ lies in ℓ^2 , while $(\varepsilon_j)_j$ does not. Hence $(\langle h, v_j + d_j \rangle)_j \notin \ell^2$, so the new family is not even Bessel.

Now suppose $\varepsilon \in \ell^2(J)$ and $\delta = \|\varepsilon\|_2 \geq \sqrt{A}$. On $\ell^2(J)$, take the frame $v_j = \sqrt{A}e_j$. Choose a unit vector y with $|\langle y, e_j \rangle| = \varepsilon_j/\delta$, let P_y be the orthogonal projection onto $\mathbb{F}y$, and put

$$d_j = -P_y v_j.$$

Then $\|d_j\| = \sqrt{A}\varepsilon_j/\delta \leq \varepsilon_j$, but $v_j + d_j \in y^\perp$ for every j . The perturbed family is not complete and hence is not a frame. This proves necessity, including the boundary $\delta = \sqrt{A}$. $\square$

6 Checks, upgrade attempts, and scope

Algebra audit. All four central assertions reduce to operator identities: $MSM^* = D$, $WW^* = D$, $S^{-1}SS^{-1} = S^{-1}$, and the norm bound $\|E\| \leq \|\varepsilon\|_2$. No compactness, spectral discreteness, or finite-dimensional matrix argument is used.

Deep upgrades. The first attempt gave the finite-dimensional formulas with a unitary factor. The first upgrade identified coisometries as the correct infinite-dimensional parameter and covered arbitrary countable frames. The second upgrade extracted the sharp canonical-dual bound. The third upgrade replaced a one-sided small-perturbation estimate by the exact minimax threshold in Theorem 5, including counterexamples both outside ℓ^2 and at the boundary. Corollary 3 gives a fully explicit diagonalizing displacement, but no claim is made that it minimizes distance to the full nonconvex class of diagonal-frame-operator frames.

Novelty check. Bounded primary-source searches through 11 August 2026 covered the source title and the phrases *diagonal frame operator*, *frame perturbation*, $MSM^* = D$, and coisometric parametrization. The searches located the source and related work on prescribed/perturbed frame operators, but no paper explicitly answering the two displayed source questions by Theorems 1 and 2. The factorization is elementary enough that it may be implicit in standard operator-frame theory; novelty confidence is therefore moderate rather than high.

Human-review recommendation. Check that coisometry, rather than unitarity, is necessary in infinite dimensions; verify the orthogonal-row equivalence; and audit the two necessity constructions in Theorem 5.

References

- [1] O. Asipchuk, J. Glidewell, and L. Rodriguez, *Additive Stability of Frames*, [arXiv:2309.06331](#).
- [2] P. Massey and M. Ruiz, *Minimization of convex functionals over frame operators*; [arXiv:0710.1258](#).